

Introduction to Statistical Methods

(S1-26_AIMLCZC418) - Assignment 1

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Q1. Travel Time Analysis

Data: 42, 45, 47, 44, 46, 48, 43, 45, 47, 85
 $n = 10$

(a) Mean and Median

Formula: Mean $\bar{x} = \sum x_i / n$

$$\sum x_i = 42 + 45 + 47 + 44 + 46 + 48 + 43 + 45 + 47 + 85 = 492$$

$$\bar{x} = 492 / 10 = 49.2 \text{ minutes}$$

Formula: Median (even n) = average of $(n/2)$ th and $(n/2 + 1)$ th terms in sorted data.

Sorted: 42, 43, 44, 45, 45, 46, 47, 47, 48, 85

$n = 10$, so median = avg of 5th and 6th terms = $(45 + 46) / 2$
 $= 45.5 \text{ minutes}$

(b) IQR Method for Outliers

Formula: $Q1$ = median of lower half

$Q3$ = median of upper half

$$IQR = Q3 - Q1$$

Outlier limits: lower = $Q1 - 1.5(IQR)$,

upper = $Q3 + 1.5(IQR)$

Lower half (5 values): 42, 43, 44, 45, 45 $\rightarrow Q1$ = middle

value = 44

Upper half (5 values): 46, 47, 47, 48, 85 $\rightarrow Q3$ = middle value
 $= 47$

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$$\Rightarrow \text{IQR} = 47 - 44 = 3$$

$$\text{Lower limit} = 44 - 1.5(3) = \cancel{44 - 1.5}, 44 - 4.5 = 39.5$$

$$\text{Upper limit} = 47 + 1.5(3) = 47 + 4.5 = 51.5$$

Since $85 > 51.5 \rightarrow 85$ lies outside the upper limit, so it is an outlier.

(c) Mean vs Median - Justification:

The mean (49.2) is inflated by the outlier value 85, since mean uses every data point including extreme ones. The median (45.5) is based on position/rank, so it is unaffected by the extreme value. Since 9 of 10 days had travel time between 42-48 minutes, the median better represents the typical travel time, because it is resistant to the distortion caused by the outlier.

Q2. Probability with colored Balls

$$\text{Total Balls} = 5(\text{red}) + 4(\text{blue}) + 3(\text{green}) = 12$$

(a) $P(\text{red or blue})$

$$P(R) = 5/12, \quad P(B) = 4/12$$

$$P(R \text{ or } B) = 5/12 + 4/12 = 9/12 = 3/4.$$

(b) $P(\text{not green})$

$$\text{Formula: } P(\text{not } G) = 1 - P(G).$$

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$$P(G) = 3/12$$

$$P(\text{not } G) = 1 - 3/12 = 9/12 = 3/4.$$

(c) Mutual Exclusivity - Justification

Two events are mutually exclusive if they cannot occur at the same time (their intersection is empty: $A \cap B = \emptyset$).

Since only 1 ball is selected, it cannot be both red and blue simultaneously. Therefore, "selecting a red ball" and "selecting a blue ball" are mutually exclusive events.

Q3. Bayes' Theorem - Defective Components

Given: $P(A) = 0.70$ (Machine A)

$P(B) = 0.30$ (Machine B)

$P(D|A) = 0.02$, $P(D|B) = 0.05$

(a) Overall probability of defective (Law of Total Probability)

Formula: $P(D) = P(A) \cdot P(D|A) + P(B) \cdot P(D|B)$

$$P(D) = (0.70)(0.02) + (0.30)(0.05)$$

$$P(D) = 0.014 + 0.015$$

$$P(D) = 0.029$$

(b) $P(\text{Machine B} | \text{Defective})$ - Bayes' Theorem

Formula: $P(B|D) = \frac{P(D|B) \cdot P(B)}{P(D)}$

$$P(B|D) = \frac{(0.05) \times 0.3}{0.029} = \frac{0.015}{0.029} = 0.5172 = 51.72\%$$

Ans

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Q4. Discrete Probability Distribution

X	0	1	2	3
P(X)	0.40	0.30	0.20	0.10

(a) Validity Check:

Conditions for a valid probability distribution

1.) $0 \leq P(x_i) \leq 1$ for all i

2.) $\sum P(x_i) = 1$

Verification:

⇒ All values (0.40, 0.30, 0.20, 0.10) lie between 0 & 1.

⇒ $\sum P(x) = 0.40 + 0.30 + 0.20 + 0.10 = 1.00$ ✓

Both conditions satisfied ⇒ valid probability distribution

(b) Expected value $E(X)$:

Formula: $E(X) = \sum (x_i \cdot P(x_i))$

$E(X) = 0(0.40) + 1(0.30) + 2(0.20) + 3(0.10)$

$E(X) = 0 + 0.30 + 0.40 + 0.30$

$E(X) = 1.0$

Interpretation: On avg, a customer 1 item per visit.

(c) $P(X \geq 2)$

Formula: $P(X \geq 2) = P(X=2) + P(X=3)$

$P(X \geq 2) = 0.20 + 0.10 = 0.30$ Ans.